

ROUTE 5 (PATH 5) — COMPLETE ORGANIZER ANSWER KEY & OPERATIONS GUIDE

CONFIDENTIAL • OFFICIAL ORGANIZER MASTER MANUAL

ROUTE: COE (LIB) → AIML → CSE → MECH → AUDI

ORGANIZER INSTRUCTION: This official manual corresponds to PATH5_FINAL_ODK.xlsx. All typed answers strictly enforce **MANDATORY UPPERCASE**.

1. MASTER FLOW & VOLUNTEER PASS CODES

Stage	Location	Challenge	Variant_Count	Expected_Answer_or_Code	Time_Limit	Unlocks	Notes
Stage 1: COE (LIB)	COE Library Zone	🌿 Object — COE Board	1 (Universal)	P5-COE-PASS	3-5 mins	NEXT BLOCK: AIML & Start Code	Locate physical object -> Volunteer verifies and gives pass code
Stage 2: AIML Start	AIML Checkpoint A	Arrival at AIML Station	1 (Universal)	AIML-START	Immediate	AIML Phase 1 (Optical Illusion)	Volunteer assigns Variant A-G
Stage 3: AIML P1 Optical	AIML Checkpoint A	Optical Illusion Animal Counting	7 Variants (A-G)	12 / 11 / 10 / 9 / 8 / 7 / 6	3-4 mins	AIML Phase 2 (Missing Concept)	Reused from Route 2 (decreasing counts)
Stage 4: AIML P2 Concept	AIML Checkpoint B	Riddle of the Missing Concept	7 Variants (A-G)	REGRESSION / CLUSTERING / PRECISION / DECISION TREE / OUTLIER / GRADIENT DESCENT / OVERFITTING	4-5 mins	AIML Phase 3 (AI Image Prompt)	Reused from Route 2 dataset tables
Stage 5: AIML P3 AI Image	AIML Checkpoint C (7 Computers)	AI Image Prompt & Recreation	7 Variants (A-G)	Volunteer visual match -> P5-AIML-PASS	5-8 mins	NEXT BLOCK: CSE & Start Code	Reused from Route 2 (10s observation, verbal description, AI recreation)
Stage 6: CSE Start	CSE Checkpoint A (Floor 1 / Lab 101)	Arrival at CSE Station	1 (Universal)	CSE-START	Immediate	CSE-01 (Emoji Math Riddle)	Reused from Route 4
Stage 7: CSE-01 Emoji Math	CSE Checkpoint A (Floor 1 / Lab 101)	Emoji Arithmetic Riddle	7 Variants (A-G)	20 / 30 / 44 / 36 / 43 / 33 / 41	3-4 mins	CSE-02 (AI Sorting Machine)	Reused from Route 4
Stage 8: CSE-02 AI Sorting	CSE Checkpoint B (Floor 2 / Room 204)	AI Card Classification Rule Discovery	7 Variants (A-G)	PRIME / CONVEX / PALINDROME / SYMMETRIC / EVEN PARITY / SUPERVISED / LINEAR	4-5 mins	CSE-03 (Binary Identity)	Reused from Route 4
Stage 9: CSE-03 Binary Identity	CSE Checkpoint C (Floor 3 / Room 305)	8-bit ASCII Binary Byte Decoding	3 Variants ONLY (A, B, C)	302 / 415 / 278	3-4 mins	CSE-04 (Morse Transmission)	Reused from Route 4 (EXACTLY 3 variants)
Stage 10: CSE-04 Morse	CSE Checkpoint D (Floor 1 / Room 112)	Morse Code Audio/Visual Transmission	7 Variants (A-G)	ALGORITHM / COMPILER / FIREWALL / DATABASE / NETWORK / PROTOCOL / RECURSION	3-4 mins	CSE-05 (Lock Grid)	Reused from Route 4
Stage 11: CSE-05 Lock Grid	CSE Checkpoint E (Floor 2 / Room 218)	6 Sequential Security Locks Gauntlet	7 Variants x 6 Locks = 42 Qs	See Lock Grid Bank Sheet	7-10 mins	CSE-06 (Building-Wide QR Password Hunt)	Reused from Route 4
Stage 12: CSE-06 QR Password Hunt	CSE Multi-Floor Corridors (Floors 1, 2, 3)	Building-Wide QR Hunt + Physical Tasks + Decoys	7 Route Variants	CYBER-NEXUS-71 / QUANTUM-VECTOR-82 / SHADOW-MATRIX-93 / CRYPTO-BEACON-14 / SYNAPSE-SIGNAL-25 / BINARY-VORTEX-36 / SILICON-PULSE-47	8-12 mins	CSE Volunteer Verification (P5-CSE-PASS)	Reused from Route 4
Stage 13: MECH Sound Clue	Campus Corridor / Transition Area	Sound Challenge (mech_sound_p5.wav)	1 (Universal)	Sound: ENGINE / Block: MECH / Code: P5-MECH-START	3-4 mins	MECH Mini-Challenges	Reused from Route 1 (audio does not reveal answer in question)
Stage 14: MECH Mini-Challenges	Mechanical Engineering Workshop	Human Shape (7 poses) -> Hidden Garland (5 items) -> Bomb Defusal (knots)	3 Phases (Universal)	P5-MECH-P1 -> P5-MECH-P2 -> P5-MECH-P3 -> P5-MECH-PASS	8-12 mins	NEXT BLOCK: AUDI	Reused from Route 1 physical challenges
Stage 15: Audi 1 (Riddle)	Main Auditorium	Auditorium Riddle ("A hall of echoes, where voices rise...")	1 (Universal)	AUDITORIUM / AUDI	2-3 mins	Audi 2 (Stage Riddle)	Universal final riddle
Stage 16: Audi 2 (Stage)	Main Auditorium Stage	Stage Riddle ("Raised above the wooden floor...")	1 (Universal)	STAGE	2-3 mins	Final Physical Challenge	Universal stage riddle
Stage 17: Final Physical Trial	Grand Stage	Final Physical Coordination & Agility Challenge	1 (Universal)	Final Answer 1: STAGE / Final Answer 2: FINAL-PATH5	5-8 mins	🏆 PATH 5 COMPLETE	Judges verify trial and issue final 2 pass answers

2. AIML TRI-PHASE MASTER ANSWER BANK

Variant	AIML_P1_Animal_Count	AIML_P1_Animal_Word	AIML_P1_Description	AIML_P2_Context	AIML_P2_Concept_Answer	AIML_P2_Clue	AIML_P3_AI_Target	AIML_Volunteer_Code
A	12	TWELVE	Elephant, Lion, Wolf, Eagle, Deer, Rabbit, Fox, Bear, Owl, Swan, Dolphin, Horse	Weather and Temperature Forecasting	REGRESSION	Predicting continuous numerical target values from features	Cyberpunk Neon Hover-Car in Rain	P5-AIML-PASS
B	11	ELEVEN	Lion, Tiger, Falcon, Cheetah, Zebra, Giraffe, Monkey, Squirrel, Peacock, Fox, Otter	E-Commerce Customer Segmentation	CLUSTERING	Unsupervised grouping of data points by similarity	Golden Mechanical Steampunk Pocket Owl	P5-AIML-PASS
C	10	TEN	Elephant, Leopard, Bull, Hawk, Kangaroo, Koala, Panda, Wolf, Lynx, Camel	Medical Diagnostic Test Evaluation	PRECISION	Ratio of true positives over total positive predictions	Astronaut Botanical Greenhouse on Mars	P5-AIML-PASS
D	9	NINE	Jaguar, Eagle, Stallion, Bison,	Email Spam Detection Tree	DECISION TREE	Flowchart-like structure of	Floating Crystal Island	P5-AIML-PASS

Variant	AIML_P1_Animal_Count	AIML_P1_Animal_Word	AIML_P1_Description	AIML_P2_Context	AIML_P2_Concept_Answer	AIML_P2_Clue	AIML_P3_AI_Target	AIML_Volunteer_Code
			Antelope, Gazelle, Owl, Falcon, Badger			conditional branching rules	Castle in Clouds	
E	8	EIGHT	Tiger, Bear, Wolf, Moose, Fox, Beaver, Raccoon, Crane	Network Intrusion Detection	OUTLIER	Data point that significantly deviates from the remaining distribution	Mythical Jade Dragon Coiled around Temple	P5-AIML-PASS
F	7	SEVEN	Lion, Eagle, Wolf, Panther, Ram, Falcon, Boar	Model Optimization and Loss Minimization	GRADIENT DESCENT	Iterative optimization algorithm to minimize the loss function	Retro 80s Cyber Dolphin in Wireframe Sea	P5-AIML-PASS
G	6	SIX	Stallion, Eagle, Tiger, Wolf, Bear, Stag	Model Performance On High Variance Data	OVERFITTING	Model learns noise in training data and fails to generalize	Enchanted Forest Potion Alchemy Laboratory	P5-AIML-PASS

3. CSE 6-STAGE BUILDING GAUNTLET ANSWER BANK

Variant	CSE01_Emoji_Answer	CSE02_AI_Sort_Answer	CSE03_Binary_Answer	CSE04_Morse_Answer	CSE05_Lock6_Master_Code	CSE06_QR_Password	CSE_Volunteer_Code
A	20	PRIME	302	ALGORITHM	64-31-AND-STACK-STAR-71	CYBER-NEXUS-71	P5-CSE-PASS
B	30	CONVEX	415	COMPILER	21-42-OR-QUEUE-RING-82	QUANTUM-VECTOR-82	P5-CSE-PASS
C	44	PALINDROME	278	FIREWALL	49-51-NOT-ARRAY-BUS-93	SHADOW-MATRIX-93	P5-CSE-PASS
D	36	SYMMETRIC	302	DATABASE	125-60-XOR-GRAPH-MESH-14	CRYPTO-BEACON-14	P5-CSE-PASS
E	43	EVEN PARITY	415	NETWORK	19-75-NAND-TREE-8-25	SYNAPSE-SIGNAL-25	P5-CSE-PASS
F	33	SUPERVISED	278	PROTOCOL	28-80-NOR-HEAP-2-36	BINARY-VORTEX-36	P5-CSE-PASS
G	41	LINEAR	302	RECURSION	63-100-XNOR-LIST-LOG N-47	SILICON-PULSE-47	P5-CSE-PASS

4. CSE LOCK GRID GAUNTLET (ALL 42 QUESTIONS ACROSS 7 SETS)

Variant	Lock_Number	Lock_Question	Expected_Answer	Case_Rule
A	1	Lock 1: Arithmetic sequence: 2, 4, 8, 16, 32, [?]	64	UPPERCASE ONLY
A	2	Lock 2: Hexadecimal conversion: What is hex 0x1F in decimal?	31	UPPERCASE ONLY
A	3	Lock 3: Logic gate: What logic gate outputs 1 ONLY when both inputs are 1?	AND	UPPERCASE ONLY
A	4	Lock 4: Anagram: Unscramble the data structure: H T K C A S	STACK	UPPERCASE ONLY
A	5	Lock 5: Network topology: Which topology connects all nodes to a single central hub/switch?	STAR	UPPERCASE ONLY
A	6	Lock 6: Master Code: Combine [64, 31, AND, STACK, STAR] + PIN 71	64-31-AND-STACK-STAR-71	UPPERCASE ONLY
B	1	Lock 1: Fibonacci sequence: 1, 1, 2, 3, 5, 8, 13, [?]	21	UPPERCASE ONLY
B	2	Lock 2: Hexadecimal conversion: What is hex 0x2A in decimal?	42	UPPERCASE ONLY
B	3	Lock 3: Logic gate: What logic gate outputs 0 ONLY when both inputs are 0?	OR	UPPERCASE ONLY
B	4	Lock 4: Anagram: Unscramble the data structure: E E U Q U	QUEUE	UPPERCASE ONLY
B	5	Lock 5: Network topology: Which topology connects nodes in a closed continuous loop?	RING	UPPERCASE ONLY
B	6	Lock 6: Master Code: Combine [21, 42, OR, QUEUE, RING] + PIN 82	21-42-OR-QUEUE-RING-82	UPPERCASE ONLY
C	1	Lock 1: Square numbers: 1, 4, 9, 16, 25, 36, [?]	49	UPPERCASE ONLY
C	2	Lock 2: Hexadecimal conversion: What is hex 0x33 in decimal?	51	UPPERCASE ONLY
C	3	Lock 3: Logic gate: What gate inverts its single input bit?	NOT	UPPERCASE ONLY
C	4	Lock 4: Anagram: Unscramble the data structure: A R Y R A	ARRAY	UPPERCASE ONLY
C	5	Lock 5: Network topology: Which topology connects all devices to a single shared backbone cable?	BUS	UPPERCASE ONLY
C	6	Lock 6: Master Code: Combine [49, 51, NOT, ARRAY, BUS] + PIN 93	49-51-NOT-ARRAY-BUS-93	UPPERCASE ONLY
D	1	Lock 1: Cube numbers: 1, 8, 27, 64, [?]	125	UPPERCASE ONLY
D	2	Lock 2: Hexadecimal conversion: What is hex 0x3C in decimal?	60	UPPERCASE ONLY
D	3	Lock 3: Logic gate: What gate outputs 1 only when inputs differ (odd parity)?	XOR	UPPERCASE ONLY
D	4	Lock 4: Anagram: Unscramble the data structure: H P A G R	GRAPH	UPPERCASE ONLY
D	5	Lock 5: Network topology: Which topology connects every node directly to every other node?	MESH	UPPERCASE ONLY
D	6	Lock 6: Master Code: Combine [125, 60, XOR, GRAPH, MESH] + PIN 14	125-60-XOR-GRAPH-MESH-14	UPPERCASE ONLY
E	1	Lock 1: Prime sequence: 2, 3, 5, 7, 11, 13, 17, [?]	19	UPPERCASE ONLY
E	2	Lock 2: Hexadecimal conversion: What is hex 0x4B in decimal?	75	UPPERCASE ONLY
E	3	Lock 3: Logic gate: What universal gate is an AND followed by NOT?	NAND	UPPERCASE ONLY
E	4	Lock 4: Anagram: Unscramble the hierarchical data structure: E E R T	TREE	UPPERCASE ONLY
E	5	Lock 5: Architecture: In standard binary byte units, how many bits equal one Byte?	8	UPPERCASE ONLY
E	6	Lock 6: Master Code: Combine [19, 75, NAND, TREE, 8] + PIN 25	19-75-NAND-TREE-8-25	UPPERCASE ONLY
F	1	Lock 1: Triangular numbers: 1, 3, 6, 10, 15, 21, [?]	28	UPPERCASE ONLY
F	2	Lock 2: Hexadecimal conversion: What is hex 0x50 in decimal?	80	UPPERCASE ONLY
F	3	Lock 3: Logic gate: What universal gate is an OR followed by NOT?	NOR	UPPERCASE ONLY
F	4	Lock 4: Anagram: Unscramble the memory data structure: P E H A	HEAP	UPPERCASE ONLY
F	5	Lock 5: Architecture: How many nibbles are in a single 8-bit byte?	2	UPPERCASE ONLY
F	6	Lock 6: Master Code: Combine [28, 80, NOR, HEAP, 2] + PIN 36	28-80-NOR-HEAP-2-36	UPPERCASE ONLY
G	1	Lock 1: Powers of two minus one (Mersenne): 1, 3, 7, 15, 31, [?]	63	UPPERCASE ONLY
G	2	Lock 2: Hexadecimal conversion: What is hex 0x64 in decimal?	100	UPPERCASE ONLY
G	3	Lock 3: Logic gate: What gate outputs 1 only when both inputs are equal?	XNOR	UPPERCASE ONLY
G	4	Lock 4: Anagram: Unscramble the dynamic structure: S T I L	LIST	UPPERCASE ONLY
G	5	Lock 5: Computer science: What is the time complexity of binary search on sorted array (O notation base)?	LOG N	UPPERCASE ONLY
G	6	Lock 6: Master Code: Combine [63, 100, XNOR, LIST, LOG N] + PIN 47	63-100-XNOR-LIST-LOG N-47	UPPERCASE ONLY

5. MECH SOUND & MINI-CHALLENGES MASTER KEY

Step	Question_or_Task	Media_File	Accepted_Answers	Pass_or_Start_Code	Notes
Step 1: Audio Acoustic Clue	Listen to audio recording (mech_sound_p5.wav) and identify the mechanical machine/component	mech_sound_p5.wav	IC ENGINE / ENGINE / FOUR STROKE ENGINE	Unlocks Block Guess	Audio player inside form
Step 2: Department Block Guess	Which block does this sound direct your team to?	N/A	MECH / MECHANICAL / MECHANICAL BLOCK	Unlocks Arrival Code Input	Directs teams to MECH Workshop
Step 3: MECH Arrival Clearance	Report to MECH Workshop station volunteer	N/A	P5-MECH-START	P5-MECH-START	Issued upon team arrival at Workshop Bay
Step 4: Mini-Game Phase 1 (Human Shape)	4 team members recreate all 7 synchronized human poses sequentially (mech_human_shape.png)	mech_human_shape.png	P5-MECH-P1	P5-MECH-P1	Volunteer verifies pose accuracy
Step 5: Mini-Game Phase 2 (Hidden Garland)	Find 5 hidden objects across the challenge area and bind into a garland using rope (mech_hidden_garland.png)	mech_hidden_garland.png	P5-MECH-P2	P5-MECH-P2	Volunteer inspects bound garland
Step 6: Mini-Game Phase 3 (Bomb Defusal)	Carefully untangle and remove designated knot sequence from the challenge puzzle box (mech_bomb_defusal.png)	mech_bomb_defusal.png	P5-MECH-P3	P5-MECH-P3	Volunteer inspects defused puzzle box
Step 7: MECH Final Block Clearance	Present completed trials to Head MECH Marshall	N/A	P5-MECH-PASS	P5-MECH-PASS	Qualifies team to proceed to Main Auditorium

6. DETAILED CHALLENGE DESCRIPTIONS, MATERIALS & PARALLEL STATION CAPACITIES

Stage 1: COE (Library Annex) Object Search

PARALLEL STATIONS: 3 DESKS

Event & Mechanics: 25 teams locate physical COE Board in Library Annex and present to volunteer desk.

Location: Library Annex / COE Room
Capacity: 25 Teams (3-5 min each)

Materials: COE Board object, tokens
Pass Code: P5-COE-PASS

Staffing: 3 Volunteers
Next Block: AIML Block

Stage 2: AIML Tri-Phase Computer Lab Challenge

PARALLEL STATIONS: 7 WORKSTATIONS

Event & Mechanics: Optical illusion count (12 to 6), Dataset Concept Table, and 10s AI Image Generation Prompting on lab terminals.

Location: AIML Computer Lab
Capacity: 25 Teams (Top 15 qualify)

Materials: 7 Logged-in AI prompt PCs
Pass Code: P5-AIML-PASS

Staffing: 4 Lab Volunteers
Next Block: CSE Block

Stage 3: CSE 6-Stage Building Gauntlet

PARALLEL STATIONS: 6 FLOOR STATIONS

Event & Mechanics: 15 teams complete Emoji math, AI sorting, Binary identity (3 tracks A, B, C), Morse transmission, 42 lock grid, and QR hunt.

Location: CSE Building (Floors 1-3)
Capacity: 15 Teams (Top 7 qualify)

Materials: QR code cards, lock stations
Pass Code: P5-CSE-PASS

Staffing: 6 CSE Marshalls
Next Block: MECH Block

Stage 4: MECH Sound Clue & Mini-Challenges

PARALLEL STATIONS: 6 WORKSHOP BAYS

Event & Mechanics: 7 teams identify audio clue (ENGINE → MECH) and complete Human Shape, Hidden Garland, and Bomb Defusal knots.

Location: MECH Workshop Bay
Capacity: 7 Teams (Top 5 qualify)

Materials: Pose mats, garland parts, knot boxes
Pass Code: P5-MECH-PASS

Staffing: 6 Bay Judges + 1 Head Judge
Next Block: Main Auditorium

Stage 5: Main Auditorium Grand Finale

PARALLEL STATIONS: 2 STAGE ARENAS

Event & Mechanics: Finalists solve Auditorium riddle (AUDITORIUM), Stage riddle (STAGE), and complete live physical coordination trial.

Location: Main Auditorium Stage
Final Codes: Ans 1: STAGE • Ans 2: FINAL-PATH5

Materials: Coordination challenge props
Conclusion: 🏆 PATH 5 COMPLETE

Staffing: 3 Chief Judges
Podium: 1st, 2nd, 3rd Place Award Ceremony